

5. Development Process

Hello!

While we wait for everyone,
feel free to share in chat what
your favorite emoji is!

Feedback

:)

Clear explanations

Examples and visuals

Interaction

Questions get answered

:/

Pacing is too fast

Slides aren't the same in Codio

Different way of interaction (polls,
asking if anyone has answer)

SEO Tech
Developer

Review

Databases

What?

Place to store structured data

Often use SQL programming language to create tables, get and store data

Why?

Apps need to permanently store, organize, and quickly retrieve data securely, especially when data is large, shared, constantly changing

Multiple users at same time, data consistency / rules

Manage relationships between connected data

Imagine!

Have data

- CSV file
- retrieved from API
- etc

How?



Put data into
database to store

What database?

SQLite3

Imagine!

Have data

- CSV file
- retrieved from API
- etc



Load into code
using Python
and pandas

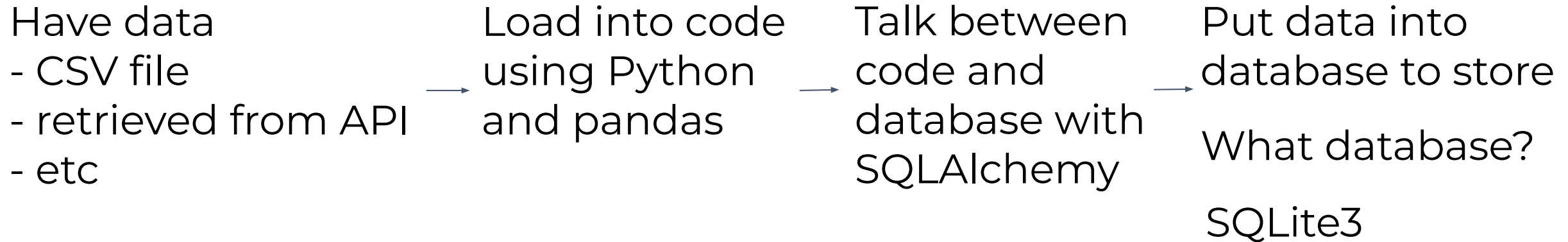


Put data into
database to store

What database?

SQLite3

Imagine!



Development Process Overview

Imagine

Have app idea

What's next?

Development process



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Design Documents

Design Considerations

Two important ways to think about what you are building:

Usefulness

What problem to solve?

Who would use finished product?

What is **smallest useful piece** you can build?

Other features that add value?

Technical

Data inputs?

User output?

How do **inputs become outputs?**

What **technology** to use (e.g. API, database, libraries)?

Design Document - Pitch

First step of project: create buy in often “pitch deck” (short slides) with:

- Project name
- What problem are you solving? (in about 2 sentences)
- Who / What does project interface with? (e.g. people, systems, hardware)
- Inputs?
- Outputs?
- List 5 steps to go from input -> output
- Biggest risk?
- Define success

HINT HINT Useful for presentations over course of residency!

Design Document - Specification

After successful pitch, (usually) **Product Manager** writes specification / requirements document

Contains same elements as pitch but with **more explanations**

Might additionally include testing plan / other analytics like expected number of customers to use feature

Why?

Everyone (managers, software engineers, designers, product managers) are **all on same page**, know what to build and why

Feedback and iteration for improvements

Design documents

What?

Written document to **detail project** for usefulness and used technology

Why?

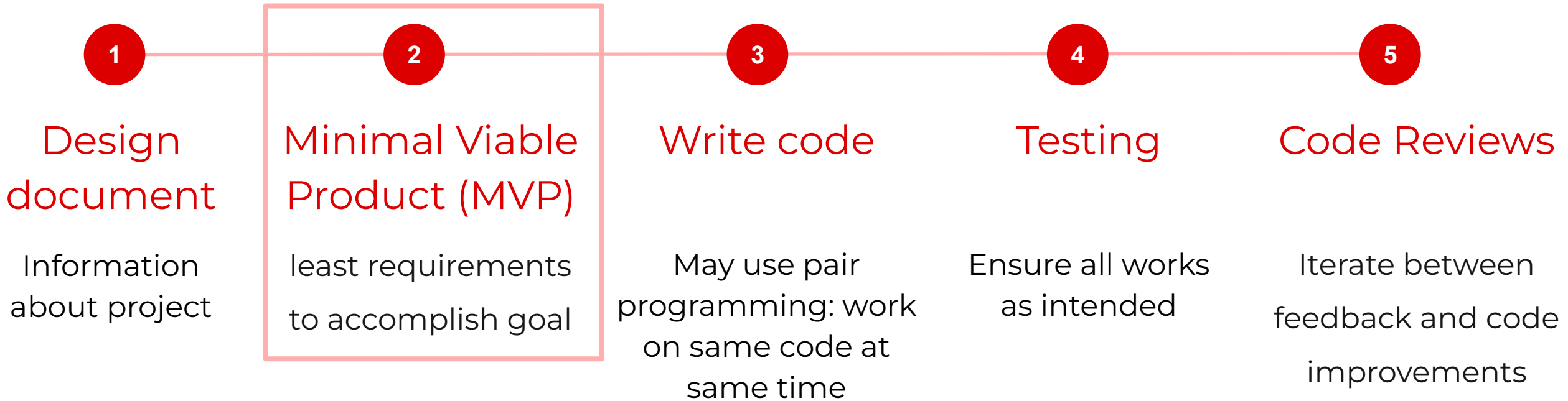
Everyone (managers, software engineers, designers, product managers) are **all on same page**, know what to build and why

Feedback and iteration for improvements

Development process



Development process



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MVP

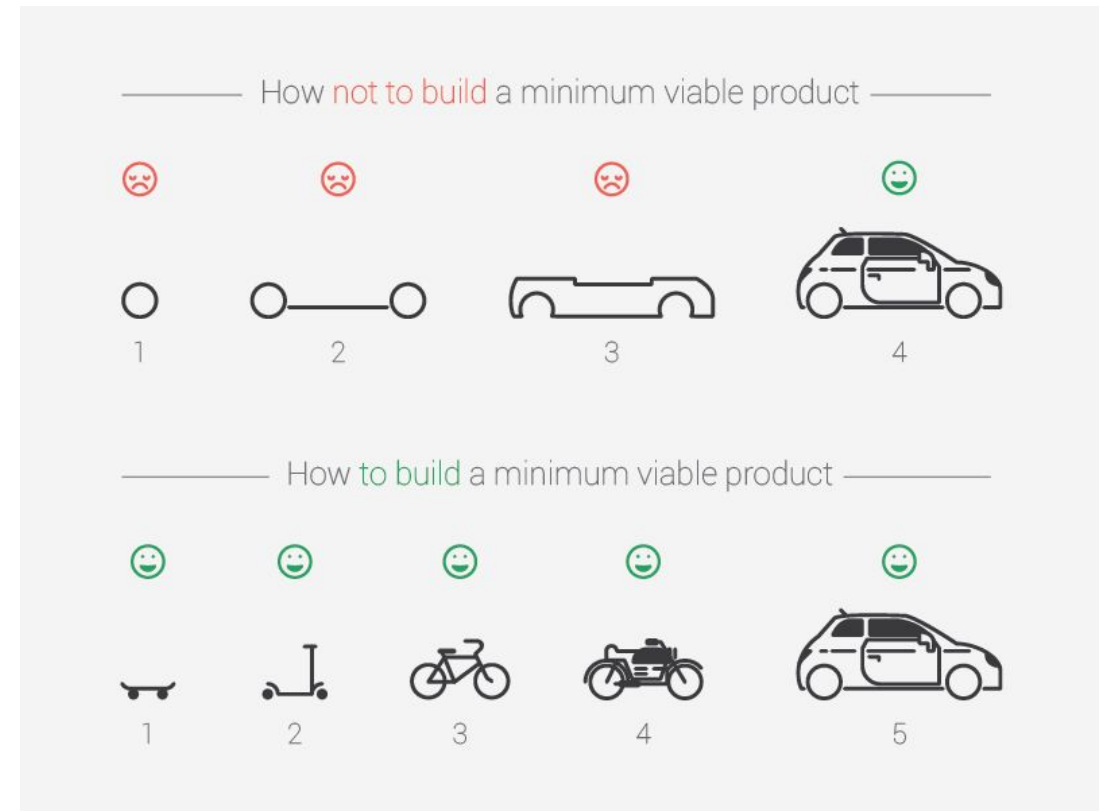
Minimum Viable Product

Absolute minimum requirements to solve problem

Simplest version of working app

Get basic app working

Then build more features



Why?

Usually want to include infinite number of features (distinct piece of functionality) - scope creep (rabbit hole)

Can't simultaneously build infinite number of features

Choose most important features that can get app launched to world

Sooner give to users, sooner get feedback, iterate

Question: what is **absolutely needed** in order to solve the problem our app is trying to solve?

Everything else is “nice to have” (aka build later)

Example

Problem to solve: connect students with same major at a university

What is the minimum needed to solve this problem?

- Create account / have profile, login functionality
- Find other students with same major
- Contact other students

Nice to have: everything else

MVP

What?

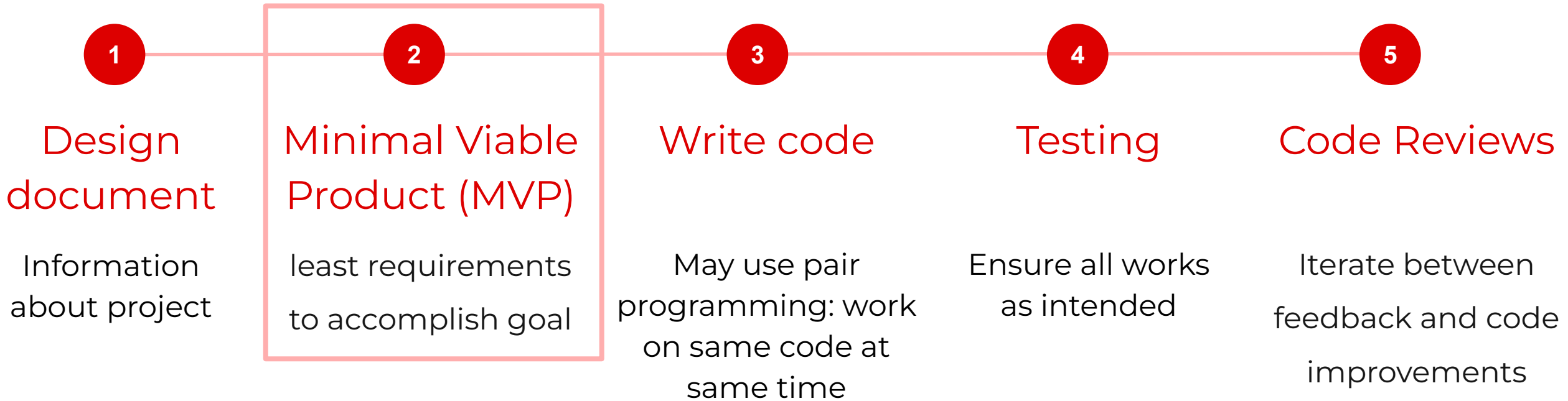
Absolute minimum requirements to solve the problem that app is trying to solve

Why?

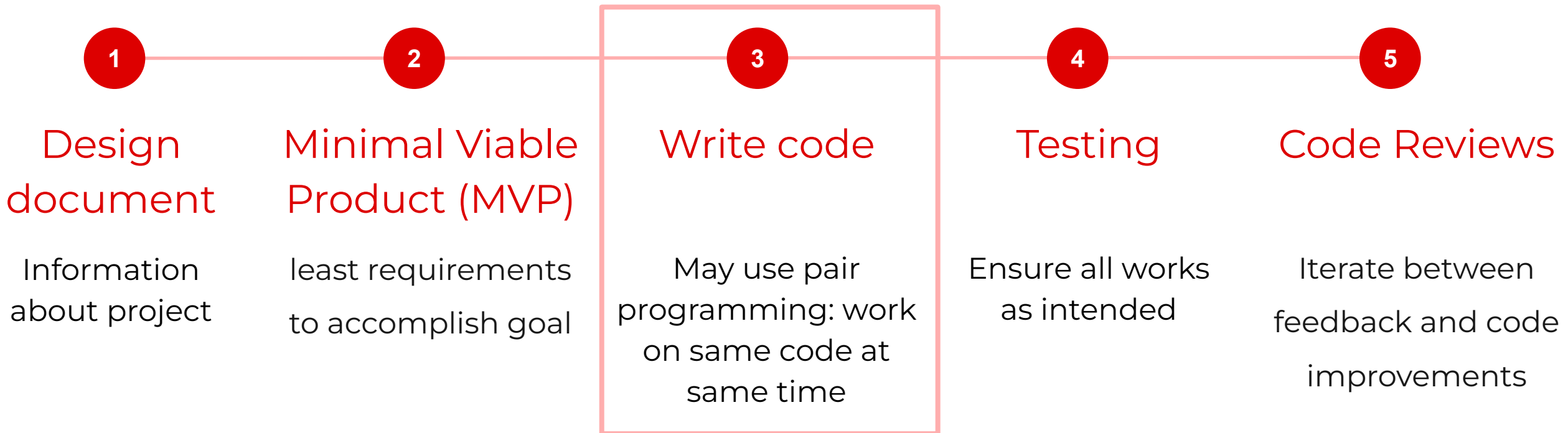
Prevent scope creep

Get app out to users to get feedback / improve while build more features

Development process



Development process



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Pair Programming

What is pair programming?

Two developers work on same piece of code at same time

One developer types and offers thoughts / comments - **driver**

Other developer checks for mistakes, look up documentation, guide driving - **navigator**

Why?

Improve code quality, speed up learning, reduce mistakes - especially for complex work

Fewer bugs - problems caught immediately instead of during code review / testing

Knowledge sharing - learn from each other, knowledge is spread out

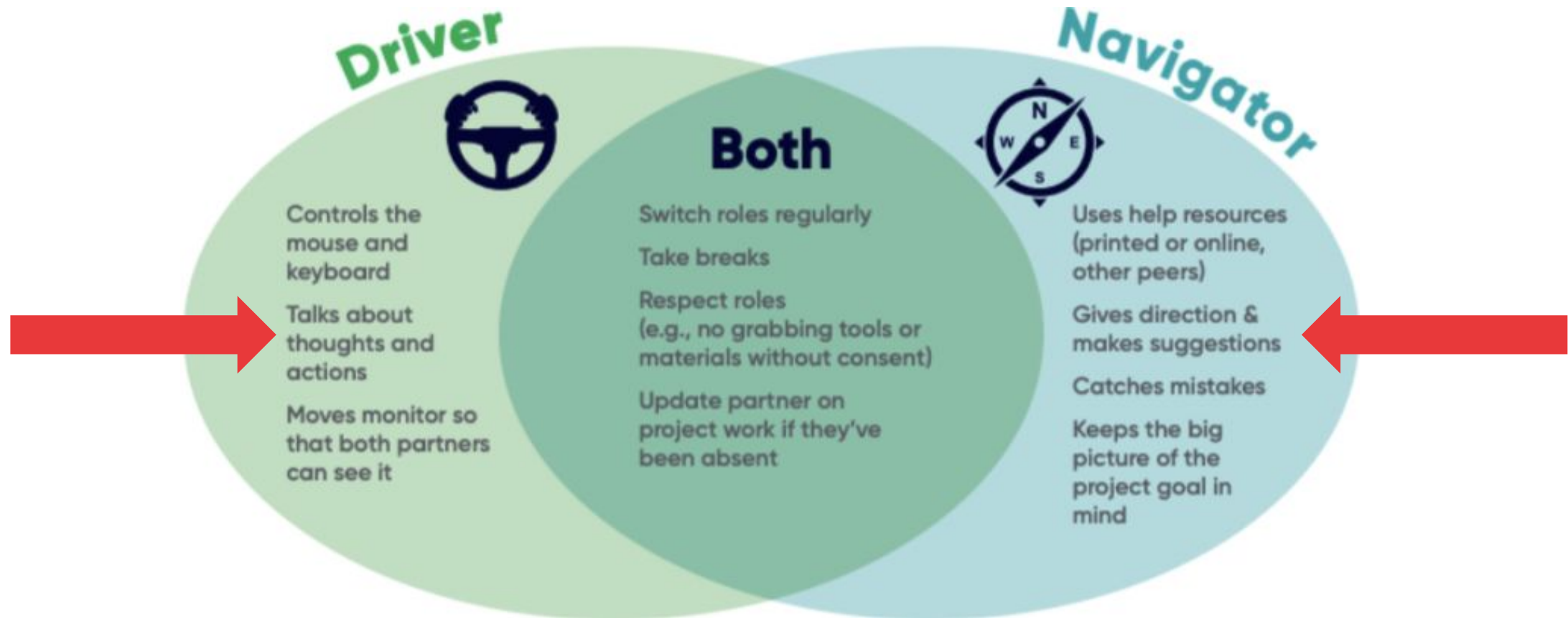
Faster onboarding - new team members are productive more quickly

Reduced "getting stuck" time - two heads better than one!

Higher confidence in correctness - code feedback during development

Pair programming is
NOT
taking turns typing
quietly in corner!

What is Pair Programming?



5. Design Documents and Paired Programming

Pair programming

What?

Two developers work on same piece of code at same time

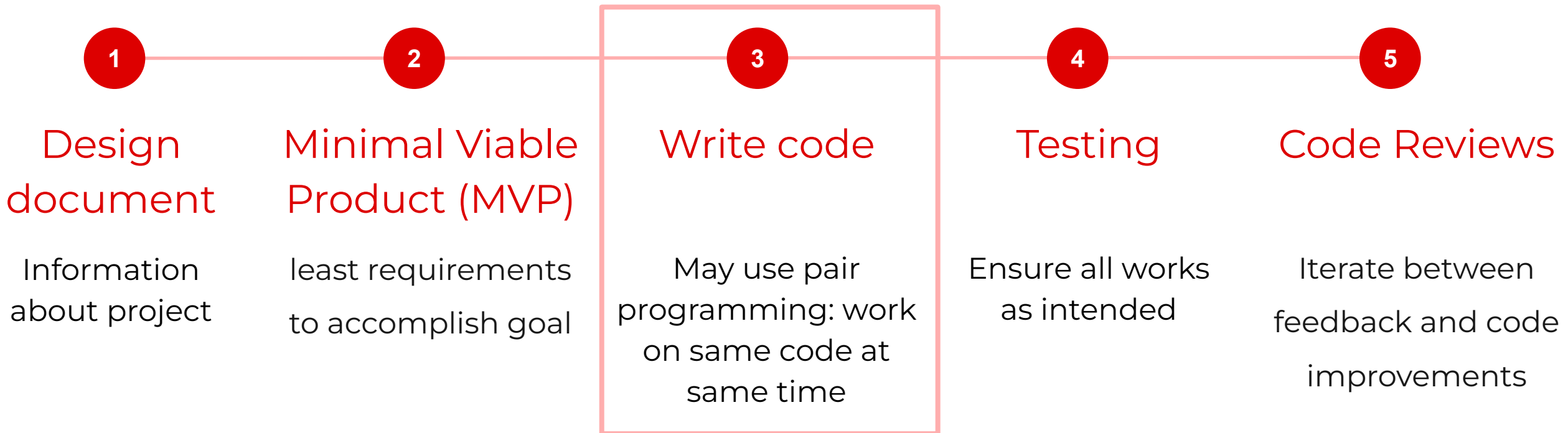
Driver and Navigator

Why?

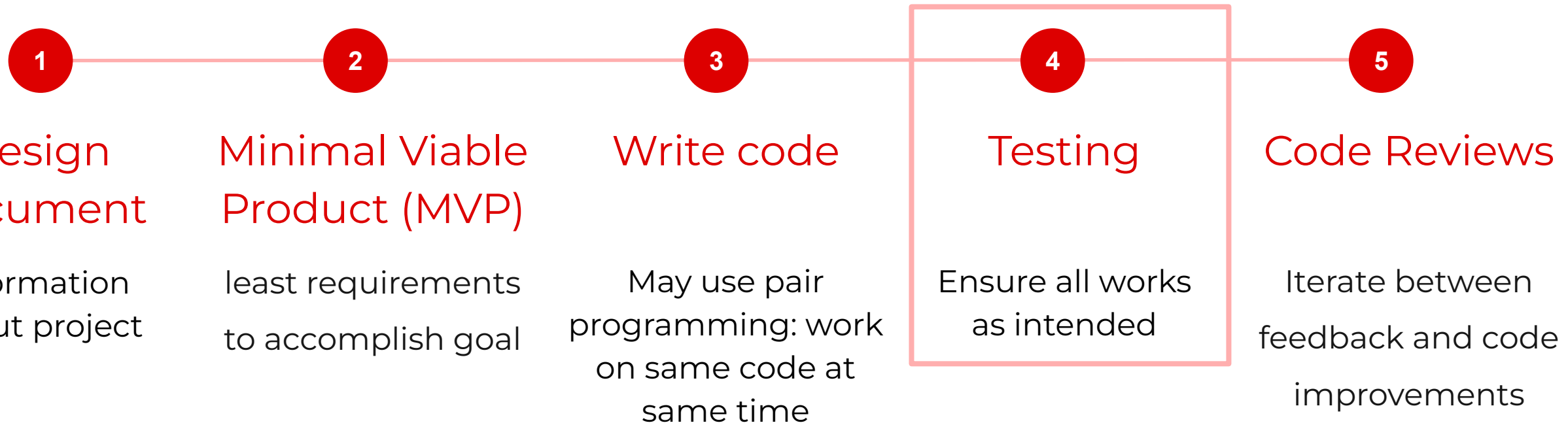
Improve code quality, speed up learning, reduce mistakes - especially for complex work

Fewer bugs, knowledge sharing, faster onboarding, reduced “getting stuck” time, higher confidence in correctness

Development process



Development process



Testing

What?

Write function that does certain task

How to know it's correct?

Try giving different inputs -> see if correct output

Manual: using `input()` or `print()` to check with own eyes

Manual testing is slow: think of app like Gmail or Instagram or...??

Every time change code, need to try entire app??!!

Automate tests to check whether various inputs are giving correct outputs

After writing tests, click button and automatically get told whether code is correct!

What kinds of tests?

Unit Tests - individual functions

Integration Tests - integrated components, parts are interacting with each other

System Tests - entire app / system

Acceptance Tests - done by client to ensure meets needs

When and Who Writes Tests?

Developer writes tests for any new code **before** committing

Some organizations have dedicated test engineers (QA / Quality Assurance engineer): create and run tests

Code structure

Modules for reusability, ensure discrete, testable function

As break code into methods, think about (write in comments) test cases for each

Example: if method includes conditional, test cases “cover” all branches

Unit Testing

Unit Tests

Verify expected behavior of functions

Useful for checking complicated functions (but in reality, all functions)

UnitTest – Example

```
def absolute_value(num):  
→ if num < 0:  
    answer = num * -1  
→ else:  
    answer = num  
return answer
```

What should get tested?

In this example, check every conditional block

What types of numbers should get tested?

How? unittest – Python Library

Use `assert` methods:

- `assertEqual(a, b)` - “make sure a and b are equal”
- `assertNotEqual(a, b)`
- `assertTrue(a)`
- `assertFalse(a)`
- `assertRaises(a)`
- `assertAlmostEqual(a, b)` - “make sure a - b to 7th decimal point is 0”
- `assertNotAlmostEqual(a, b)`
- `assertIn(a, list_b)`

Throws error when function returns `False` - quiet otherwise

UnitTest – Example

```
import unittest
from myMath import add, absolute_value # or module

class TestFileName(unittest.TestCase):

    def test_add(self):
        self.assertEqual(add(2, 1), 3)
        self.assertEqual(add(2.1, 1.2), 3.3)

    def test_absolute_value(self):
        self.assertEqual(absolute_value(1), 1)
        self.assertEqual(absolute_value(-5), 5)
        self.assertEqual(absolute_value(0), 0)
```

Automating Unit Tests

GitHub Actions

- Automates tasks
- Specified in YAML (Yet Another Markup Language)
- White space matters

Create

`.github/workflows/tests.yaml`

- Name workflow
- Install Python and unittest
- Run unittest

```
# tests.yaml
name: Tests
on: push

jobs:
  check-tests:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Setup python
        uses: actions/setup-python@v2
        with:
          python-version: 3.11.3
      - name: Install tools
        run: python -m pip install
              --upgrade pip pytest
      - name: Test with unittest
        run: python3 -m unittest test.py
```

Dependencies

Real software depends many parts:

- database
- API
- file systems
- email services
- payment gateways
- message queues
- much, much more!

Systems are constantly changing - not good for tests!

Dependencies: example

`order_service.place_order(order)`

Uses database, payment provider, email service

Test might:

- Fail when internet is down
- Fail because payment provider is unavailable
- Send real emails
- Require complex setup and cleanup

How to fix this?

Test doubles

Think of stunt double in movies: look and act like real thing

Objects or functions to stand in for real parts

Types:

Fake - code pretending to be fully implemented code (mock or stub)

Mock - fake object that mimics real, verifies interactions with dependencies

Stub - code represents unimplemented object, returns predefined responses

- never fail unit test but mock can
- could be replaced when functionality is added

Dummy - object passed around but never used (parameter required, test doesn't use)

Mocks

Use mocks to see:

- When functions were called
- How many times specific functions were called
- What arguments were passed to function when called

Useful for controlling code behavior

Example: function updates user's Instagram account. When testing, don't want to change person's Instagram account

Create mock of account and use dummy for changing during testing

How?

`unittest.mock` library provides us with:

`Mock()` class - creates fake object (with functions / attributes) that you can control / inspect

`patch()` methods - temporarily replace real object with fake one

Mock() example

Suppose code uses external service (calls public API, database, etc)

```
def greet(user_service, user_id):  
    user = user_service.get_user(user_id)  
    return f"Hello {user['name']}"
```

user_service calls an external service (pretend this is the case)

Shouldn't actually call service (why? might induce costs, service down, slow, etc)

Use Mock() for service (user_service in above example)

Mock() example

```
from unittest.mock import Mock

def test_greet():
    fake_service = Mock() # create object
    # Assign function and return value
    fake_service.get_user.return_value = {"name": "Alice"}
    result = greet(fake_service, 123) # call function
    # Make sure function call returns correctly
    assert result == "Hello Alice"
    fake_service.get_user.assert_called_once_with(123) # inspect
```

Patch() example

Suppose code uses external service but service isn't function parameter

```
def greet(user_id):  
    user_service = UserService() # creates own UserService  
    user = user_service.get_user(user_id)  
    return f"Hello {user['name']}"
```

UserService isn't accessible outside function

Why not just use it? Same reasons as before - UserService class might call real service which could incur costs, be down, be slow, etc

Use Patch() for it

Patch() example

```
from unittest.mock import patch
from app import greet

@patch("app.UserService") # Which file and object to patch
def test_greet(mock_user_service_class):
    # Configure instance returned by UserService()
    mock_instance = mock_user_service_class.return_value
    mock_instance.get_user.return_value = { "name": "Alice" }
    result = greet(123)

    assert result == "Hello Alice"
    mock_user_service_class.assert_called_once_with()
    mock_instance.get_user.assert_called_once_with(123)
```

Stubbing

2 use cases:

1. Need classes / methods to exist for syntactical correctness but don't need to do anything. Use `pass` keyword

```
def future_function(*args):  
    pass
```

2. To-be-implemented code needs to do something. Since know test cases, write minimal code needed to pass tests. Usually, series of conditionals:

```
def fibonacci(n):  
    if n == 1:  
        return 1  
    elif n == 6:  
        return 8  
    else:  
        return 233
```

Mock vs Stub in Python

Stub: provides predetermined value (ie when `get_user` returns, always return `{"name": "Alice"}`)

Mock: records interactions for verifications (ie how many times function called, what arguments function was called with, etc)

Didn't we just see both in `Mock()`?

Yep. `Mock()` in Python does both

Mock() example

```
from unittest.mock import Mock

def test_greet():
    fake_service = Mock() # create object
    # Assign function and return value stub
    fake_service.get_user.return_value = {"name": "Alice"}
    result = greet(fake_service, 123) # call function
    # Make sure function call returns correctly
    assert result == "Hello Alice" mock
    fake_service.get_user.assert_called_once_with(123) # inspect
```

Test Code Coverage

What is Test Code Coverage?

Metric to measure if testing is sufficient

Which lines of codes are executed when run tests

Pros:

- Easy to automate
- Clear, quantitative indicator

Cons:

- Limited usefulness in terms of test case coverage (particularly for exceptions)

Coverage.py

Docs: <https://coverage.readthedocs.io/en/latest/>

If not installed: `pip3 install coverage`

To run: `python3 -m coverage run -m unittest test_file_name.py`

Generate report, to view with command line: `python3 -m coverage report`

Pretty Reports

Transform report into HTML page: `python3 -m coverage html`

View in Codio by right-clicking **index.html** > **Preview Static**

Click on file name to see details

The screenshot shows the Codio IDE interface. On the left, the 'Filetree' sidebar displays the project structure under 'EDEITRICK'. The 'Test Code Coverage (master)' folder is expanded, showing subfolders like '.guides', 'python-style-checker-examp', and 'htmlcov'. The 'index.html' file is selected. On the right, the 'index.html' file is open in the editor, displaying a 'Coverage report: 94%' summary. Below the summary is a table with columns: 'Module', 'statements', 'missing', 'excluded', and 'coverage'. The table lists 'badStyle.py' (100% coverage) and 'test.py' (92% coverage), with a 'Total' of 94% coverage. At the bottom of the report, it says 'coverage.py v5.5, created at 2021-06-25 13:53 +0000'. A red arrow points to the 'Preview static' button in the context menu that appears when right-clicking on 'index.html'.

Module	statements	missing	excluded	coverage
badStyle.py	5	0	0	100%
test.py	12	1	0	92%
Total	17	1	0	94%

coverage.py v5.5, created at 2021-06-25 13:53 +0000

Testing

What?

Writing code to test code **automatically**

Can use **test doubles** (fakes, mocks, stubs, dummies) to remove dependencies while testing

Why?

Ensure code is **working as intended**

Don't need to manually test every time code is changed

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Development Pipeline

Development Pipeline

When code is pushed, doesn't automatically go to users

What do you think might happen?

Development Pipeline

Automated run of static analysis tools:

- style checker
- bug checker
- security checker

Automated run of tests

Code review by another developer

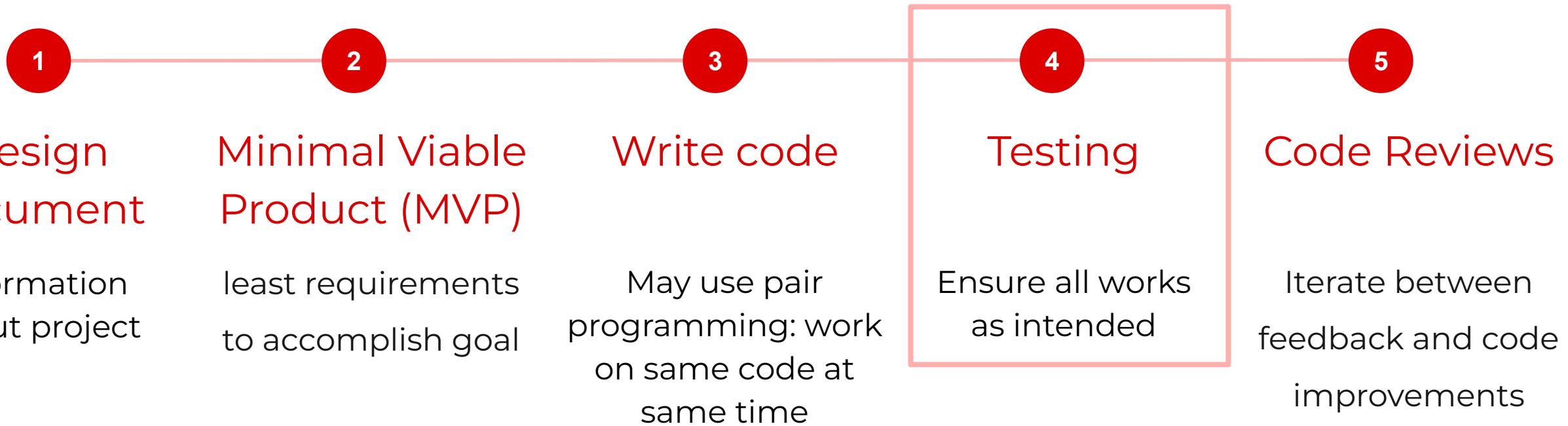
Manual test by QA (Quality Assurance)

Automation

Computers consistently run programs against code, instead of manually checking

Create programs (or markup files) to verify behavior of other programs (or even check styling) for every added / edited file

Development process

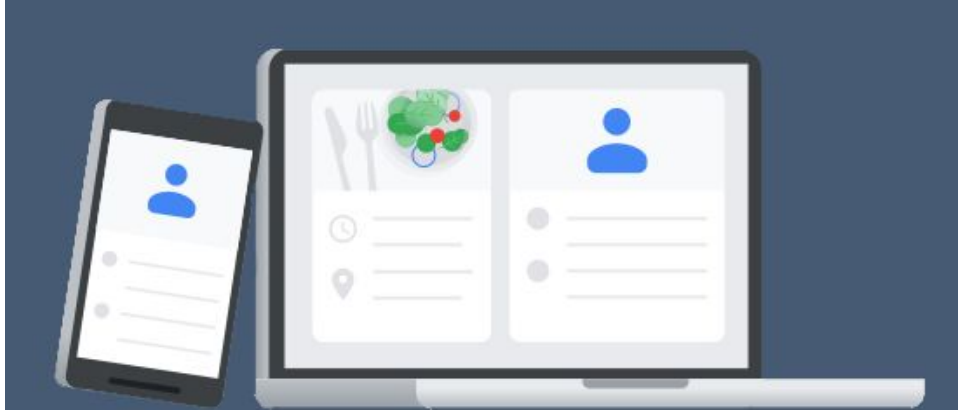


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Types of programmers

End-to-end applications

App (mobile or web)



Open page

Might need data that changes over time

Ask server



Server

Database



Get data back



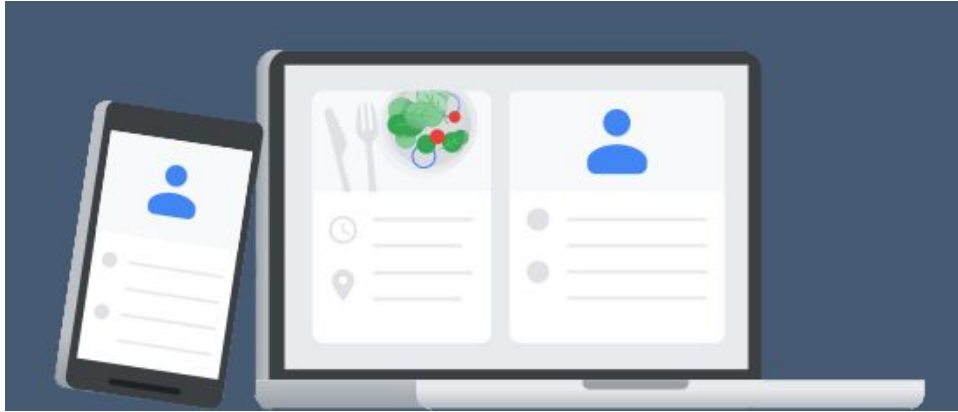
Ask database



Figures out what data to get

End-to-end applications

App (mobile or web)



App shows data

Database



Open page

Might need data that changes over time

Ask server

Returns data to app

Get data back

Ask database

Gathers data together



Server

Lots of work!

Visuals

Server

Database

APIs

Security

Scalability

...and more!

Types of programmers

Different programmers take responsibility for different parts of app

Visuals / user experience (how users interact with product): **frontend**

Server, databases, APIs, scalability: **backend**

Software engineers: **frontend engineer**, **backend engineer**, **fullstack engineer**

Fullstack: does frontend and backend work

Backend, Back-End, or Back End?

It depends!

Adjective: backend / back-end (ie backend engineer)

Noun: back end (“The back end of the system handles data processing”)

Same goes for frontend and fullstack

Most of the time use backend

Front End vs. Back End vs. Full Stack?

During this program, could be helpful to reflect on what kind of work you enjoy doing

Why?

determine what kind of internship you want

how to shape your portfolio to show off particular skills

identify skills you might want to build next!

Week 2 Project Overview

Built in **pairs** - assigned partner within EM group

Use all skills from Week 1:

- API integration
- Creation, Querying, and Updating of Databases
- Adherence to PEP8 style (coming soon)
- Testing plan and appropriate unit tests
- GitHub Automation for style checkers and unit tests (coming soon)

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Thank you!